Demand Uncertainty, Capital Specificity, and Industry Evolution*

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Abstract

The existence of capital that is specific to an industry (its investment cost exceeds its value-in-use in other industries) radically alters the equilibrium in an otherwise straightforward neoclassical model. In particular, as demand falls exit is impeded and as demand rises the entry of new capital is also impeded in the sense that a positive rate of profit at a given point in time need not induce investment. This paper investigates the interplay between capital mobility, demand uncertainty, industry investment, and price. In particular, the existence and uniqueness of a competitive equilibrium in a stationary environment is established. The equilibrium is fully characterized by two thresholds $L \leq H$: entry occurs when the ratio of intrinsic demand to capital is above H and exit occurs when this ratio is below L . Whether or not exit is reversible (it is not when the act of exit renders the capital unsuitable for future use in the industry), equilibrium demands that L increase and H decrease with mobility.

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Introduction

The equalization of rates of return expressed by Adam Smith's invisible hand metaphor and driven by the movement of various factors to higher valued uses has, quite properly, occupied centre stage in economic analyses. In the textbook model of perfect competition, short-term equilibration arises from the adjustment of production rates and long-term equilibration is brought about by the flow of mobile capital from unproductive to productive uses. In particular, an unexpected upward shift in the demand function induces entry of new capacity or firms: entry proceeds until profits fall to zero. Conversely, downward shifts in the demand function are followed by the exit of capacity or firms; again, industry profits rise to the zero-profit equilibrium condition. However, if capital is specific to an industry, having a value-in-use outside industries in which capital is specific are frequently subject to sustained periods

of overcapacity and 'cut-throat' competition. An excellent example is the oil-tanker industry and its upstream supplier, the tanker construction industry. Prior to 1973, the supply and demand for world oil-tanker capacity were in balance and both had expanded at an average rate of 12% per annum. The unexpected tripling of oil prices by OPEC in 1973 inaugurated a new era of diminished growth in oil transportation. Between 1973 and 1978, shipments of oil by tanker grew at only 4.2% per year while available tanker capacity grew at 10.2% per year. The growth in excess capacity occurred because operators continued to (incorrectly) anticipate an upturn in oil shipping. Because tankers are specific to the industry, the capacity imbalance persisted well into the 1980s with disastrous effects on industry profits. [1]

Sustained overcapacity has also been a characteristic of the world steel, aluminium, and, more recently, major parts of the petrochemical industries. For example, a 5 -year period of ethylene overcapacity in Japan ended in 1987, sparking investments in new capacity by Japanese producers and speculative investments by new Korean chemical firms. As a result, total Japanese and Korean capacity is now projected to be 10.7 million tons per year in 1993 compared with a government-projected demand of 8.8 million tons per year. [2] In this type of situation, engineering economists often point out that the profits on the 'upside'-if demand is much higher than expected-more than compensate for the losses if demand is at expected or lower than expected levels. The problem with this simple option story is that it does not take into account the fact that the upside scenario will last only until new capacity or new sources of imports arrive, whereas the downside scenario can last for a decade or more.

The key feature of this setting-demand uncertainty coupled with industry specific capital—is that myopic entry (and exit) policies are no longer optimal. Even risk-neutral atomistic price-taking firms must adopt entry and exit policies that depend upon the nature of demand uncertainty and on the impact of similar policy decisions by other firms. The frequency with which industrialists are surprised at the immobility of extant capacity suggests that this situation resists a simple analysis. A classic example is the bituminous coal industry. In this industry immobility of capacity is aggravated because water seeps into mines that are not operated. Hence, it is more economic to operate a mine at a loss than to close it for an extended period. As Cohen and Cyert (1965, p. 154) noted, "It has been the object of much legislation aimed at maintaining profits in the industry. As we shall see, this inability to reach equilibrium arises because exit is difficult whereas entry is relatively easy.'

This paper develops a formal model of capital specificity and investigates the nature of the optimal entry and exit policy. Apart from establishing the existence of an equilibrium, the analysis focuses on the impact of increases in uncertainty and decreases in mobility, where increases in uncertainty correspond to mean preserving increases in risk and mobility is measured by the ratio of investment cost to value in next-best use. In addition, we offer a numeric example which provides an estimate of the magnitude of the wedge between the rate of return triggering myopic entry and that associated with optimal entry. The example also sheds light on the magnitude of the uncertainty faced by new capital with the optimal entry policy.

[1] For details see "The Oil Tanker Shipping Industry,' in Porter (1983). [2] 'Far East Ethylene: Glut Ahead,' Chemical Week, August 8, 1990, p. 21.

Industry Specific Capital

The notions of irreversible investment, capital specificity, and sunk cost are closely intertwined. Expenditures are irreversible whenever the item produced (be it a capital good, goodwill, brand name capital, or the formation of a firm) cannor be returned costlessly to its former physical state. For example, the milling of metals and the production of drill presses are irreversible investments.

These irreversible investments are specialized to a single use whenever their value in that use exceeds their value in alternative uses or when there are costs associated with transferring the asset to an alternative use. The full productive value of a drill press can be extracted through its employment in any one of a large number of industrial uses. On the other hand, while a railroad boxcar might have some value when used as a warehouse or even as a restaurant, one anticipates that its maximum value will be attained only when it is used as rolling stock. Consequently, the railroad car is industryspecific but the drill-press is not. Specificity does not imply, nor is it caused by, impediments to resale: there are ready markets for both rolling stock and drill presses. Specificity is associated with the uniqueness of an asset's use, not of its user.

To be concrete, we define an investment's salvageability s as the ratio of its value in its next-to-best use to its investment cost, and define specificity as $1 - s$.

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Just as the concept of a resource's cost takes its meaning from the context of available decisions about alternative uses, so too the specificity of an investment depends upon the range and associated values of alternative uses. For example, an asset can be specific to a particular production technology, with its specificity measured by the relative reduction in its value were this technology unavailable. Similarly, an asset has a specificity with respect to an entire industry, measured by comparing its value in-use within the industry to its value in its best use outside the industry. Thus, assets can have specificities with respect to particular plants, firms, products, nations, workgroups, and customers.

Sunk costs are the ex post consequences of investments in specific capital: the portion of investment cost that is sunk is its specificity. It is useful to note that sunk costs, like specificity, have meaning only with respect to a decision context; frequently it is helpful to speak of costs being sunk with respect to a particular use, rather than as simply 'sunk'. Thus, when Baumol et al. (1982, p. 282) suggest that airlines have low levels of sunk costs, they apparently mean that route and/or firm specificity is low. But investments in aircraft are at the same time strongly specific to the air transportation industry: the level of aircraft cost sunk with respect to the industry is quite high. [3]

In this paper we are concerned with capital that is specific to an industry; henceforth the only relevant decision context is the addition of new homogeneous capital to the industry and its possible withdrawal for use outside the industry.

Related Literature

Despite the obvious fact that real firms and industries employ substantial quantities of highly specialized capital, the implications for the theory of the firm and for industrial organization have only begun to be explored. One stream of work on this subject, pioneered by Williamson (1975) and by Klein, et al. (1978), has focused on the problems of opportunism associated with the sale of the services of specialized assets. When an asset's specialization is to a particular user, so that its investment cost cannot be fully recovered from other users or employments, its specificity with respect to that user is potentially appropriable through opportunistic bargaining or recontracting. The transaction costs accompanying the sale of the services of user-specific assets, it is argued, induce vertical integration or other forms of non-market control.

Spence (1979), Dixit (1980), Eaton and Lipsey (1980), and Baumol et al. (1982) pursue another line and see sunk costs as creating barriers to entry which an incumbent might strategically manipulate to deter entry. According to this view, the owners of unspecialized capital cannot credibly deter the entry of new capital into an industry because their own exit is, essentially, costless. Conversely, the wedge between the investment cost and salvage value of industry-specific capital-called an exit barrier-can induce a corresponding entry barrier.

A third realm of inquiry concerns the impact of immobile capital on macroeconomic equilibria. McLure (1970, 1971) and Fullerron (1983) examine sectorial investment responses to a one-time unanticipated change in tax policy. In a more complex model, Grossman and Shapiro (1982) investigate the acquisition by workers of generalized versus specific (immobile) skills when facing uncertainty about the future demand for skills.

A fourth important stream of thought examines the impact of capital specificity on investment decisions. Early work by Arrow (1968) established the non-myopic nature of investment decisions under certainty when investment is irreversible. A number of studies have looked at the impact of uncertainty on irreversible investments; the results are sensitive to the fine structure of the models. Smith (1969) examined a monopolist's investment decision in the face of demand uncertainty, while Hartman (1972), Zabel (1973), Nickell (1977), and Abel (1983) studied a competitive firm investing in capacity to maximize expected profits in the face of an exogenous stochastic price process. These studies clarified the connection between uncertainty and investment that is irreversible or subject to adjustment costs, but they did not examine the nature of the competitive equilibrium induced by a collective of decision-makers mutually aware of the interdependence created by the irreversibility of their investments.

Lucas and Prescott (1971) inaugurated a fifth stream by stepping beyond partial equilibrium analysis. They constructed a model in which investment decisions and market prices are jointly determined. In their model, firms have zero production costs but face (proportional) capital adjustment costs which impede the rapid entry of new capital. In contrast to Lucas and Prescott, our model has no adjustment costs or other impediments hindering the inflow of new capital. Instead, investment is naturally limited by the inability to fully re-

[4] This distinction is important co mortgage lenders, among others. The fact that aircraft are casily transferred to other routes or firms does nor protect the lender against a decline in the prospects of the industry as a whole. During the 1981 post-deregulation recession, many banks found that their collateral, although freely marketable, commanded market prices below the mortgage amounts.

cover investment costs upon exit: declines in demand without the parallel exit of specialized capital may produce profit streams too small to recover the cost of investment. As in Lucas and Prescott, we assume constant returns to scale and stationarity of the stochastic process generating shifts in demand. Our specification of this stochastic process is less general but yields a simple and precise characterization of the competitive equilibrium.

Lambson's (1988) and Dixit's (1991a,b) work is contemporaneous with our own. Dixit analyses a specialized case of the model presented here. As we do, Dixit posits industry-specific capital, a multiplicitive stochastic process that affects the demand function, constant returns to scale, and solves for the equilibrium in which investment is endogenous. However, Dixit's specification is less general. Our model assumes a general demand function, a homogeneous production function, and non-degenerate demand shocks; Dixit's model specifies a constant elasticity of demand, a Cobb-Douglass production function, and geometric Brownian demand shocks. [4] The more restrictive assumptions allow him to obtain a closed form solution for the critical ratio of demand to capital which induces new investment.

In the next section the concept of capital specificity is discussed, the details of the model are spelled out, and comparative statics results are obtained. In section 3 the exit of partially mobile capital is introduced. The case of reversible exit is easily treated with the results already in hand from section 2, but the case of irreversible exit requires a more delicate analysis to establish the existence and uniqueness of a memoryless competitive equilibrium. Comparisons amongst these three versions of the model-two with exit and one without exit are presented in section 4.

The Basic Model

The two state variables in our discrete time model are X_n, parameterizing the underlying demand at time n, and K_n, the total quantity of industry specific capital invested in the industry by time n. New capital enters the industry immediately after the demand X_n has been announced but prior to production in period n. As will be made explicit our model contains no economies of scale as well as sufficient stationarity so that attention can be focused upon the ratio $R_n = X_n/K_n$ of demand to capital rather than the two state variables. This permits collapsing the two state variables into one.

To obtain the appropriate scaling of costs so as to exclude scale economies and to ensure the homogeneity of firms, let A be any strictly increasing cost function. The cost $C(q,k)$ of producing q with k units of capital-which does not include the opportunity cost of the k units of capital-is given by

$$C(q,k) = kA(q/k), \quad q \geq 0, k > 0.$$

We assume that A' is strictly increasing and continuous so that it has a continuous strictly increasing inverse labelled S. Our assumption that A is convex is needed to ensure that the industry is fully characterized by the sum of the firms' capital stocks-so costs are unaffected by the division of capital amongst the individual firms. Cobb-Douglas production

[4] The corresponding restriction in our discrete-time model is that the demand shock is lognormal.

functions$A(x) = bx^n$ with $n > 1$-as well as logarithmic, exponential, power, and products and (non-negative) sums of such functions (e.g., $1 + xe^x + \ln(1 + x)$) are allowed.

Each firm acts as a risk-neutral price taker, maximizing its one-period operating profit by setting $p = C'(q, k) = A'(q, k)$ and producing

$$q = kS(p).$$

Summing (2) over the firms in the industry gives the industry supply curve $KS(p)$, where K is the total capital in the industry.

In order to model an industry demand function shifting stochastically in time, we posit a fundamental demand curve $D(p)$, sometimes written simply as D, and a sequence of independent random variables $\langle Y$ ». The demand curve D_n at time n is given by

$$D_n = X_n D,$$

where ($X_0 = 1$)

$$X_n = X_{n-1} Y_n = \prod_{i=1}^{n} Y_i, \quad n = 1, 2, \ldots$$

The function D is assumed to be continuous and strictly decreasing. Stationarity is attained by assuming that each Y_n is distributed as the nondegenerate random variable Y. To avoid trivialities assume

$$P(Y \leq 0) = 0, P(Y > 1) > 0, \text{ and } P(Y < 1) > 0.$$

If demand is X_n and the capital stock is K_n, then the industry demand curve is $X_n D(p)$ and the industry supply curve is $K_n S(p)$. Equating supply and demand gives $K_n S(p) = X_n D(p)$. It is clear that the unique market clearing price which solves this equation depends only upon the ratio $r = X_n/K_n$ of intrinsic demand to installed capacity. [5] Furthermore, the market clearing price p_r is a strictly increasing and continuous function of r because increases in r are equivalent to increases in demand. Denote by $\pi(r)$ the one-period operating profit [6] accruing to each unit of capital when the industry's demand to capital ratio is r. Of course,

$$\pi(r) = p_r S(p_r) - A(S(p_r)).$$

Because p_r solves $p = A'(S(p))$, it is easy to see that $\pi'(r) = p_r' S(p_r) > 0$. Consequently.

$$\pi(r) \text{ is continuous and strictly increasing in } r.$$

We assume, without loss of generality, that the price of capital is \$1 per unit. Firms, acting rationally, will purchase capital in order to enter the industry if and only if the expected discounted value associated with an investment of K units of capital is at least K. Because there is free entry and no private information, the net expected value of entry will be at most zero.

[5] The symbol r suppresses the period n to highlight the fact that price depends only upon the value of the ratio $X_\#/K_\#$. The symbol $R_\#$ is employed whenever probabilistic aspects or remporal identifications are particularly important.

Although the model proceeds as if capital does not deteriorate, exponential decay of the capital stock is easily incorporated. To do so let $\gamma < 1$ be the fraction of the capital stock remaining intact at the end of each period. The increase in the ratio from r to r/γ caused by the decay can be obtained merely by replacing each demand shift Y_n by Y_n/γ.

Irreversible Entry

The fundamental issue addressed in this paper is the nature of the equilibrium resulting from impediments to capital mobility. Unlike Lucas and Prescott's adjustment cost, our model contains no entry barriers. We begin by analysing the case in which capital is totally immobile: exit is not permitted. [7] Two versions of the model in which capital is partially mobile are considered in section 3.

When acting as price takers, the firms in this model are homogeneous with respect to scale so that their size distribution is immaterial and entry can take place at any scale. This fungibility and homogeneity, the fact that $\pi(r)$ is strictly increasing, and the Markovian and stationary nature of demand suggests that the rate of entry will increase with r. Moreover, given r, we anticipate current entry behaviour will be independent of the past history of both demand and entry. The simplest strategy of this variety is one we term a reservation ratio policy. Such a policy is characterized by a single parameter, label it ξ, as follows: enter if and only if $r > \xi$, where r is the current value of the ratio X_n/K_n. Of course r will never exceed ξ when the atomistic agents follow a reservation ratio policy with parameter ξ. To see this recall that capital enters in period n immediately after demand is announced and just before production occurs. An increase in demand which-except for entry of new capital-would raise the ratio above ξ will immediately induce enough entry to reduce the ratio to precisely ξ. Thus, while the ratio will never exceed ξ, it equals ξ in each period which witnesses entry.

A consequence of $r \leq \xi$ and (7) is the fact that at the time of entry the oneperiod operating profit is at its maximum. Let $\beta < 1$ denote the one-period discount factor so $\pi(\xi) > 1 - \beta$, else the present value of operating profits would be less than the cost of entry.

We will now verify that there is a unique ξ such that the value of entry is negative whenever $r < \xi$ and zero for $r \geq \xi$. Toward this end define $U(r;\xi)$ to be the expected discounted return (operating profit) over the infinite time horizon associated with the investment of one unit of capital when the current value of the demand to capital ratio is r and the agents are all investing in accord with a reservation ratio policy with parameter ξ.[8] It is clear upon reflection that U satisfies"

$$U(r;\xi) = \pi(r) + \beta EU(rY \wedge \xi;\xi), \quad r \leq \xi.$$

Lemma 1. The return function $U(r;\xi)$ is continuous and strictly increasing in r and ξ. In

" We distinguish between operating profit, which does not include a charge for capital, and net profit, which does. Given capital immobility and uncertainty, the appropriate charge for capital's use is the subject of our analysis and cannot be set a priori. [7] If $\pi(0) < (0)$, then some units of capital would elect to exit when the ratio r of demand to capital is small, even if the salvage values were zero. In the current version of the model, however, the immobility of capital precludes exit. Thus (as is the case for mines and nuclear power plants) free disposal is not allowed.

particular, g is a continuous and strictly increasing function, where g is defined by $g(\xi) = U(\xi, \xi)$, for $\xi > 0$. The proofs of all Lemmas and Theorems have been placed in the Appendix to enhance continuity.

It is immediate from Lemma 1 that if

$$\underline{L} \equiv \lim_{\xi \to 0'} g(\xi) < 1 \text{ and } \bar{L} \equiv \lim_{\xi \to x} g(\xi) > 1,$$

then there is a unique solution to $g(\xi) = 1$; label this solution ξ. As is shown below, (10) is a sufficient condition for (9). If all agents utilize the reservation ratio ξ_*, then capital enters the industry if and only if $r \geq \xi$. Consequencly, $U(r; \xi_\bullet) < 1$ whenever $r < \xi$. and $U(r; \xi.) = 1$ whenever $r \geq \xi\cdot$. Hence, the reservation ratio ξ. induces an equilibrium in this competitive industry with free-entry.

It only remains to introduce some feasibility assumptions sufficient to ensure $L < 1 < \bar{L}$. This is easily accomplished as follows. The function π is said to be feasible (with respect to the discount factor β) if there are numbers $\underline{r}$ and $\bar{r}$ such that

$$\pi(\underline{r}) < 1 - \beta < \pi(\bar{r}), \quad 0 < \underline{r} < \bar{r} < \infty.$$

Since $1 - \beta$ is the 'normal' return on capital, this condition is clearly innocuous.

Theorem 1. If π is feasible, then $\xi\cdot$ is the only reservation ratio that induces a competitive equilibrium. Furthermore, $\xi\cdot$ is the unique memoryless competitive equilibrium.

For the moment suppose, as per Dixit (1991a), that price is not allowed (say by a regulator) to rise above $\hat{p}$ and $\hat{p} < p_\xi$. Then $A(r)$, rather than $\pi(r)$, represents the one-period operating profit where

$$\hat{\pi}(r) = \min\{\pi(r), \hat{\pi}\}$$

and $\pi = \hat{p}S(\hat{p}) - A(S(\hat{p}))$ is the operating profit at maximum price $\hat{p}$. This price limitation acts as a barrier to entry in that the ratio triggering entry of new capital increases.

Corollary 1. If $\hat{p} < p_\xi, \pi(\cdot)$ is feasible, and ξ denotes the unique solution to $\hat{U}(\xi, \xi) = 1$ where $\hat{U}$ satisfies (8) with $\hat{\pi}(r)$ replacing $\pi(r)$, then $\xi > \xi$.

Increases in Uncertainty

Theorem 1 establishes the existence and uniqueness of a memoryless competitive equilibrium completely characterized by the number ξ. Next we investigate the impact of increased uncertainty. The level of demand uncertainty will be indicated by the parameter σ (which is not intended to denote the random variable's standard deviation); accordingly, the random variable Y is replaced by $Y_{"}$ and the capital entry ratio ξ. is written as ξ_σ to explicitly note its dependence on σ. An increase in the level of uncertainty-equivalently, an increase in σ - is taken to be a mean-preserving increase in risk.

We will assume that π is (strictly) concave in order to verify that an increase in uncertainty leads to a (strict) increase in ξ_σ. As demonstrated in Example 1, the strict concavity is eminently reasonable.

[κ] As the price of capital is \$1 per unit, it is clear that $U(r, \xi)$ is Tobin's q. " $x \wedge y \equiv \min\{x, y\}$.

Example 1: Suppose the elasticity of demand is constant and equal to $-b$ so that D is given by $D(p) = p^b$ for some $b < 0$ and let the cost of production be given by [10]

$$C(q,k) = k + \frac{1}{1+a^q} q^{1+al} k^u, \quad a > 0.$$

Then $p = \frac{d}{dq} C(q,k) = (q/k)^a$ so that $q = kp^{1/a}$. Equating supply and demand - $Kp^{1/a} = Xp^b$- yields the market clearing price

$$p_r = r^{a/(1-ab)}$$

and the profit function

$$\pi(r) = p_r p_r^{1/a} - \left\{ \frac{1}{1+a} \left(p_r^{1/a}\right)^{1+a} + 1 \right\} \quad = \frac{a}{1+a} p_r^{1+1/a} - 1 = \frac{a}{1+a} r^{(a+1)(1-ab)} - 1.$$

Of particular interest is the fact that π is strictly concave if (and only if) demand is elastic (i.e., $b < -1$).

This example sheds light on the effect of increases in uncertainty on firm profits. Consider a simple two period model. In the first period capital is committed to the industry, and in the second uncertainty is resolved and competitive production takes place. One common approach to modelling 'competitive' firms is to attach uncertainty directly to price and to assume that the production decisions of firms in the (competitive) industry do not affect price. In Example 1π is a convex function of price so, ceterus paribus, an increase in price uncertainty leads to an increase in $E\pi$. This result has been obtained by Hartman (1972) and Nickell (1977). By contrast, if uncertainty attaches to r (through uncertainty in the level of demand), increases in uncertainty reduce the value of $E\pi$. Thus, the impact of uncertainty on expected profit depends sharply upon whether the feedback effect on price of production decisions is modelled. In this paper we explicitly consider this feedback: price uncertainty is the joint outcome of demand uncertainty and optimal investment decisions.

Theorem 2. If π is strictly concave, then the equilibrium reservation ratio ξ_σ is a strictly increasing function of σ.

The empirical comparison between industries with differing levels of uncertainty is difficult because the state variable r is not observable. However, Theorem 2 does permit us to assert that there is a one-to-one ordering between observed profitability and demand uncertainty whenever industries are experiencing expansions of their capital stocks. This association is induced not by risk aversion but rather by the juxtaposition of uncertainty and capital specificity. Furthermore, as the return to risk-neutral capital is not a function of risk, we suggest that the volatility of profit (or price) is a surrogate for demand uncertainty. Finally, it is important for those contemplating empirical work to recognize that no such ordering is available amongst industries with stable capital stocks.

[10] This cost function is not Cobb-Douglas because the first term denotes a period expense unrelated to the rate of production.

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Some further comparative statics results are easily obtained from the model. For example, an increase in $\mu \equiv EY_\sigma$ diminishes the importance of the uncertainty so the reservation ratio ξ_μ will fall as μ increases; in fact, $\lim_{\mu \to x} \xi_\mu$ solves $\pi(\xi) = 1 - \beta$. Similarly, introducing the decay of capital at an exponential rate γ (the decay rate per period) is equivalent to increasing μ to μ/γ. Likewise an increase in the discount factor β will lead to a decrease in the equilibrium reservation ratio.

Numerical Results

In this model, the rate of return on invested capital π is greatest when $r = \xi$. so that capital is entering the industry. Because a high rate of profit is a natural stimulus to regulatory zeal, the magnitude of the entry-inducing rate of return is of interest. To investigate this issue it is necessary to assume specific forms for the cost and demand functions and a specific distribution of demand shocks. For simplicity, we use the cost and demand specifications of Example 1 and take the distribution of Y_n to be log normal- $\log(Y)$ is Normal with mean m and standard deviation θ.

Given these specifications, Monte-Carlo methods are used to compute ξ. Our approach was to define the random variable

$$T = \sum_{i=0}^{N} \beta^i \pi\left(R_i\right), \quad R_a = \xi,$$

the present value of operating profits per unit capital if the starting ratio of demand to capital is equal to the reservation ratio ξ. A series of N realizations of Y provide one realization of T, and repeated realizations of T permit us to estimate $ET \approx g(\xi)$. The single root ξ. of $g(\xi) - 1$ may be obtained by any of the standard root-finding algorithms. We also record σ_T, the standard deviation of T, as a measure of the risk facing a prospective entrant.

To calculate T we set $N = 49$. The interest rate I was set at 0.1 per period ($\beta = 1/1.1$), so that the 'normal' return on capital [11]$\pi_o = (1-\beta)/\left(1-\beta^{50}\right) = 0.09169$. We set $a = 0.2$ and studied the eighteen cases generated by taking $m = \{-0.05, 0, 0.05\}, \theta = \{0.04, 0.08, 0.12\}$, and $b = \{-0.5, -1.5\}$. Table 1 displays the calculated values of $\xi_\bullet, \pi\left(\xi_\bullet\right)/\pi_n$, and σ_T for each of these cases.

As noted in section 2.2, ξ. decreases as expected demand growth increases-this is verified by the positive relationship between ξ. and m in Table 1. In addition, we verify Theorem 2 for the case in which $\pi(r)$ is concave ($b = -1.5$) and note that the positive association between $\xi\cdot$ and θ also holds in the case in which $\pi(r)$ is convex ($b = -0.5$).

[11] $\pi_0 < /$ because our model has each unit of capital concributing profits immediately upon installation. If profies begin in the period after installation, then $\pi_n = I$.

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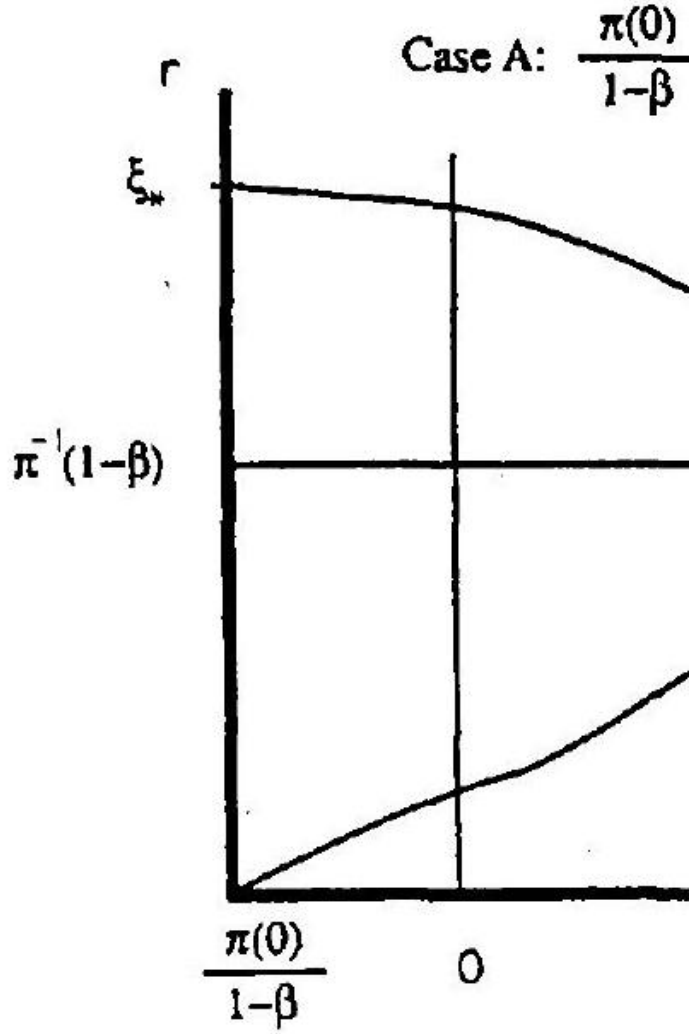
Table 1. Numerical Results

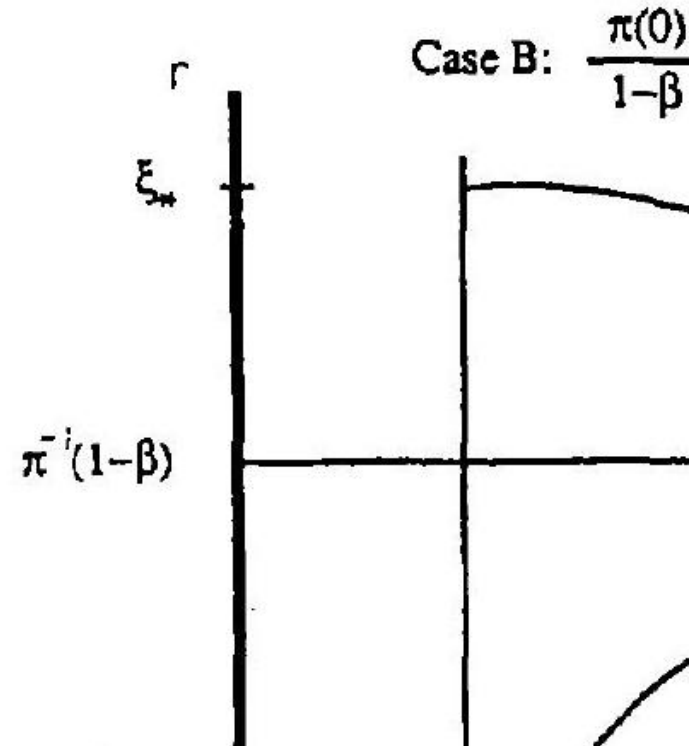
	$b = -0.5$			$b = -1.5$		
		$\boldsymbol{\theta}$		θ		
	0.04	0.08	0.12	0.04	0.08	0.12
m		Values of ξ.				
-0.05	8.19	8.24	8.35	11.35	11.43	11.64
0.00	5.97	6.32	6.67	8.16	8.66	9.13
0.05	5.61	5.72	5.91	7.68	7.83	8.10
m		Values of $\pi\left(\xi.\right)/\pi_{,}$				
-0.05	7.12	7.25	7.51	6.21	6.32	6.61
0.00	1.85	2.67	3.49	1.72	2.42	3.09
0.05	1.02	1.28	1.72	1.03	1.24	1.62
m		Values of σ_T				
-0.05	0.86	1.50	1.95	0.74	1.33	1.73
0.00	0.51	0.91	1.33	0.42	0.81	1.10
0.05	0.03	0.22	0.52	0.03	0.18	0.45

The rate-of-return premiums obtained are surprisingly large. Where growth of 5% per annum is expected ($m = 0.05$) and uncertainty is high ($\theta = 0.12$), current returns must be $1.62(b = -1.5)$ or $1.72(b = 0.5)$ times normal levels to induce entry. For industries with no expected growth $(m = 0), \pi(\xi\cdot)/\pi_0$ varies from 1.85 to 3.49 (for $b = -0.5$). That is, for capital to enter these 'no-growth' industries, the current rate of return must be two to three times the normal rate of return. If the industry is shrinking, even greater current returns must be offered to induce the creation of new industry-specific capital: Table 1 indicates that if expected growth is -5%, and the standard deviation of growth is 12%, then current returns must be 69% per period ($7:51$ times as large as normal returns) to induce entry.

A second surprising result is the magnitude of σ_T. Recall that $ET = 1$ defines the equilibrium, so that the values of σ_T shown may be interpreted as fractions of the expected present value of operating profits at the moment of investment. In a zero expected-growth industry, σ_T increases with θ from 0.51 to $1.33(b = -0.5)$. In these cases, very considerable risk is borne by new capital. In a slow growth industry ($m = 0.05$), σ_T increases with θ from 0.03 to $0.52(b = -0.5)$. Here, uncertainty dramatically increases the risk borne by new capital.

Finally, it is worth noting that when σ_T is large, it is quite difficult for managers in the in-


Case A: $\frac{\pi(0)}{1-\beta}$
r
ξ_*
$\pi^{-1}(1-\beta)$
$\frac{\pi(0)}{1-\beta}$
0


Case B: $\frac{\pi(0)}{1-\beta}$
r
ξ_*
$\pi^{-1}(1-\beta)$

only reflect risk, but also indicate the reliability of history as a predictor of the future. Were one interested in estimating ET from history, it would take 100 observations of T (5000 periods) to ensure a relative standard error of the estimate of 10% or less! This phenomena lies behind the refractory nature of the management of investment in slow-growing industries.

Partially Mobile Capital: Exit

In the model presented in section 2 capital was totally immobile and the unique equilibrium was characterized by the critical ratio ξ_σ of intrinsic demand to installed capacity below which there is no entry. In this section the basic model is extended in two nearly parallel directions so as to encompass partially mobile capital.

In both cases considered, capital is allowed to exit freely at any time without regard to the current ratio of intrinsic demand to installed capacity. In the first case each exiting unir of capital receives the lump-sum salvage payment s. Whereas exit is unhindered, it is also irreversible: the very act of exiting transforms the capital so as to prevent any future (economic) re-entry into the industry. This situation obtains, for example, when the next-best use is to sell machinery for scrap or when a railroad boxcar is converted into a restaurant.

The second case envisages not only free exit as described above, but unencumbered re-entry. Exiting capital is not destroyed; rather it stands ready to re-enter when conditions in the industry improve. When in use outside the industry, capital earns $s(1-\beta)$ per period. A possibly realistic example involves the use of a railroad boxcar as a warehouse; such use does not stifle or fetter (costless) re-entry.

Because exit is permitted the description of a competitive equilibrium mandates the use of two decision criteria, one governing exit and the other guiding entrance. The existence, uniqueness, and a comparative statics analysis of such a competitive equilibrium are developed in the ensuing subsections.

Irreversible Exit

When exit is irreversible, the state of the industry is fully described by the ratio $r = X_n/K_n$ of intrinsic demand X_n to installed capacity K_n where K_n is now the difference between the amount $E_\mu^+$of capital which has entered the industry by time n (for the first time) and the net amount E_n^-which has left the industry. Were exit reversible, the latter quantity would, by its impending re-entry, overhang the industry; accordingly, equilibrium would compel potential new entrants to know and remember E_n^-as well as X_n and K_n.

As in the case of irreversible entry, intuition leads one to anticipate that entry and exit behaviour will be independent of the past history of demand, entry, and exit. The natural extension of a reservation ratio policy is a reservation ratios policy or a two levels policy characterized by two parameters, labelled L and H, as follows: exit whenever the demand to capital ratio falls below L, enter whenever the ratio exceeds H, and leave the capital stock unchanged whenever the ratio lies berween L and H. Define $V(r; L, H)$ to be the expected discounted stream of profit associated with the investment of. one unit of capital when the current value of the demand to capital ratio is r

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and all agents enter and exit in accord with a two levels policy with parameters L and H. Then V satisfies

$$V(r; L, H) = \pi(r) + \beta EV((rY \vee L) \wedge H; L, H), \quad L \leq r \leq H.$$

Of course, $V(r; L, H)$ is defined only for values of r between L and H because values of r in excess of H induce entry and values of r below L induce exit. Moreover, it is important to recognize $V(r; L, H)$ as the expected discounted value of the operating profits associated with a unit of capital which never exits: the value of s has no influence on the value of $V(r; L, H)$. Of course, the value of s determines whether or not the two levels policy (L, H) is an equilibrium.

Whereas the return function $V(\cdot : L, H)$ associated with any given two levels policy with $L \leq H$ necessarily satisfies (12), competitive equilibrium imposes two additional constraints:

$$V(H; L, H) = 1 \quad V(L; L, H) = s.$$

The first reflects the fact that each unit of freely entering capital must earn its investment cost, whereas the second represents capital's indifference between remaining and exiting when the salvage value obtained through irreversible exit is s. Because operating costs can be negative, capital may elect to exit even if s is negative (as could be the case when there are cleanup costs).

Demonstrating that the return $V(r; L, H)$ is continuous and strictly increasing in r, L, and H is readily effected by the induction arguments exploited in section 2. Furthermore, the requisite feasibility condition (with respect to the discount factor β and salvage value s) merely requires the existence of numbers r and $\bar{r}$ such that

$$\pi(r) < (1 - \beta)s \text{ and } \pi(\vec{r}) > 1 - \beta.$$

However, the existence of an equilibrium is not at all obvious, for increasing (say) L to bring the exit condition (14) toward equilibrium impinges on the entry condition (13) and may propel it away from equilibrium. Nevertheless, feasibility as per (15) is sufficient to ensure the existence of a unique competitive equilibrium amongst the two levels policies.

Theorem 3. Let $s < 1$ be given and suppose π satisfies (15). Then there is a unique two levels policy (L_1, H_5) which satisfies (12)-(14). Furthermore, (L_5, H_s) is the unique memoryless competitive equilibrium.

In light of the particular method employed to establish the existence of a competitive equilibrium (refer to the proof of Theorem 3 in the Appendix),

discerning the impact of increases in mobility upon the competitive equilibrium is not difficult.

Theorem 4. Let $s < 1$ be given and suppose π satisfies (15). An increase in mobility induces entry and exit:

$$L'_s > 0 \quad \text{and} \quad H'_s < 0.$$

While L_s is strictly increasing and H_s is strictly decreasing in s, Figure 1 exhibits two lesser facts. First, there is no exit whenever $s \leq \pi(0)/(1-\beta)$: $L_s = 0$ for $s \leq \pi(0)/(1-\beta)$. Hence, it is clear that $H_s = \xi$. for all $s \leq \pi(0)/\ (\ 1-\beta\ )$. Second, L_s and H_s converge to $\pi^{-1}(1-\beta)$ as s increases to 1 . As expected, with perfectly mobile capital (i.e., $s = 1$) net (as opposed to operating) profits are zero in each period; with highly mobile capital (i.e., s close to 1) net operating profits are close to zero in each period.

One is tempted to conclude, falsely, that a larger value of s implies there is more capital in the industry at each point in time. (There is more capital at time zero.) What is true and does increase with s is the cumulative amount E_n^+ $[E_n^-]$of capital which has entered [exited] the industry by time n. Furthermore, the period to period variability in operating profits decreases as s increases.

Whereas it seems natural to expect a mean preserving increase in risk to retard entry if π is concave (see Theorem 2), the changes anticipated in exit behaviour are far from clear. (In fact, the changes in λ and Λ, defined in the proof of Theorem 3, are ambiguous.) We have not been able to establish an unambiguous comparative statics result. A partial explanation stems from the loss of concavity-of $V(r; L, H)$ in r-due to exit.

Reversible Exit

In this subsection capital is totally mobile. There is free entry and free exit: at any time units of capital can enter or exit. The entry cost is taken to be \$1 per unit. Upon exit the capital earns $s(1-\beta)$ per period [12] where $s \leqslant 1$. Moreover, the exiting units of capital are physically unaltered and, therefore, can elect to costlessly and instantly re-enter the industry at any time. We say the re-entry is unfettered because it is both costless and instantaneous. Because capital does not age, newly minted capital is indistinguishable from units which have exited and re-entered.

As noted earlier, equilibrium compels potential new entrants to know the net amount $E_n^-$of capital which has exited the industry by time n as well as the installed capacity K_n and the intrinsic demand X_n. But with unfertered re-entry, potential entrants will act as if the ratio r is given by $X_n/\left(K_n + E_n^-\right) = X_n/E_n^+$. The reason is simple. Per period industry profits can not exceed $s(1-\beta)$ whenever industry specific capital is employed outside the industry. When X_n/K_n exceeds $r_s \equiv \pi^{-1}(s(1-\beta))$, each unit of capital inside the industry earns more than $s(1-\beta)$. Consequently, the E_n^-extant units of industry specific capital standing ready to re-enter will in fact do so until the current per period earnings inside the industry are driven down to $s(1-\beta)$, the per period earnings in their next-best-use.

An equivalent and notationally more convenient approach is to define $\pi_s(r)$, the one-period operating profit when re-entry is unfettered and s is the salvage value ($s(1-\beta)$ is the per period earnings in the next-best-use), via

$$\pi_s(r) = \max\{s(1-\beta); \pi(r)\},$$

where r is the current value of the ratio R_n of intrinsic demand X_n to the extant supply $E_n^+$of industry specific capital. Because re-entry is instantaneous as well as costless, the exit

[12] Observe that the per period earnings $s(1-\beta)$ was selected to ensure comparability with the previous

and re-entry decisions are trivial: exit [reenter] whenever the per period earnings inside the industry fall below [exceed] $s(1-\beta)$. Accordingly, only one parameter, label it e, need be utilized to describe the memoryless strategy-termed a reservation ratio policy as in section 2-which we anticipate generates a competitive equilibrium. Using such a strategy, capital enters if and only if $r > e$ where $r = X_n/E_n^+$.

Define $W(r;e)$ to be the expected discounted return associated with the investment of one unit of capital when the current value of the ratio of demand to the extant supply of industry specific capital is r and all capital enters in accord with a reservation ratio policy with parameter e. Clearly W satisfies

$$W(r;e) = \pi_s(r) + \beta EW(rY \wedge e; e), \quad r \leq e$$

Imposing the feasibility condition (10) and replacing (8) with (18), the proof of the existence and uniqueness of a competitive equilibrium is merely the one given in section 2. Denote the unique solution to $W(e;e) = 1$ by e_s.

Theorem 5. Let $s < 1$ be given and suppose π satisfies (10). Then ε_s is the only reservation ratio that induces a competitive equilibrium. Furthermore, there are no other memoryless strategies which induce a competitive equilibrium.

The compararive statics analysis is transparent: e, decreases with s (for π_s increases with s) while the exit and re-entry point, r_s increases with s. Moreover, both e_s and r_s, converge to r_1 as s increases to 1 .

Comparisons Amongst the Three Models

Whether exit is reversible or not, capital will never elect to exit (and the feasibility condition (15) will not hold) if $s \leq \pi(0)/(1-\beta)$. Consequently, interesting comparisons amongst the equilibria emanating from the three regimes entail

$$s > \pi(0)/(1-\beta).$$

Henceforth assume this strict inequality holds. The analysis of section 3 yields

$$L_s < r_s < r_1 < H_s < \xi \ldots$$

Furthermore,

$$r_1 < e_1 < \xi\cdot$$

The first inequality is a trivial consequence of $P(Y < 1) > 0$ and $s < 1$ which imply the occurrence of ratios less than $\ell_{\cdot}$. The second inequality is verified as

model, for the discounted value of the income stream associated with the exit of a unit of capital which never re-enters (as required in subsection Irreversible Exit) is simply s.

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follows. With (19) and reversible exit, entering capital faces the one-period operating profit π_1 which is better than π, the one-period operating profit faced by immobile capital. In addition, were $e_s \geq \xi_*$, the sequence of ratios confronting the mobile capital would be (with probability 1) at least as large [exactly the same if $e_s = \xi \cdot$]. Therefore, entering mobile capital would earn a strictly larger expected discounted return; this contradicts the equilibrium entry condition. Thus, reversible exit entails an equilibrium with more willingness to enter: $e_s < \xi$.

The only comparison remaining is between H_s and e_s. We conjecture $e_s < H_s$. But the option to re-enter cuts two ways. On the one hand it imparts insurance against decreases in demand by eliminating per-period profits less than $s(1-\beta)$. On the other hand, it also effectively dampens the beneficial impact of an increase in demand. Were the ratio $1/2L_s$ and demand increased by a factor of $r_j/1/2L_j$, then the ratio with irreversible exit rises to $2r_j$; in the case of reversible exit the increase in demand calls forth re-entry, and the ratio only rises to r_s. Note that because $e_s < \xi$. and $H_1 < \xi$, the cumulative amount $E_n^+$of capital which enters the industry when exit is allowed always exceeds this amount when capital is immobile.

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Appendix

Proof of Lemma 1: We begin by defining $U_n(r;\xi)$ to be the expected discounted return associated with the investment of a unit of capital when the current value of the demand to capital ratio is r, ξ is the reservation ratio, and n periods remain. Then $(U_0(r;\xi) \equiv 0)$

$$U_n(r;\xi) = \pi(r) + \beta E U_{n-1}(rY \wedge \xi;\xi), \quad r \leq \xi, n = 1, 2, \ldots \text{ (AD)}$$

The uniform convergence of $U_n(\cdot;\xi)$ to $U(\cdot;\xi)$ on compact sets, which follows from $\beta < 1$ and π bounded on compact sets, is the key to our approach.

We claim the functions $U_n(r;\xi)$ are continuous in ξ and r, strictly increasing in r, and nondecreasing in $\xi, n = 1, 2, \ldots$ In addition, $U_n(\xi;\xi)$ is strictly increasing in ξ. By (7) we have $U_1(r;\xi) = \pi(r \wedge \xi)$ is continuous in r and ξ, strictly increasing in r, and nondecreasing in ξ. If $U_n(r;\xi)$ satisfies these properties, then $U_{n+1}(r;\xi)$ also satisfies these properties as $U_{n+1}(r;\xi)$ is merely the sum of two such functions. This verifies our claim.

Continuity of U follows from the claim and the fact that the uniform limit of continuous functions is itself continuous. Similarly, the limit of non-decreasing functions is nondecreasing. That $U(r;\xi)$ is strictly increasing in r follows from (8), $U(r;\xi)$ nondecreasing in r, and $\pi(r)$ strictly increasing.

An increase in ξ never leads to a lower ratio $R_{`j1}$. Similarly, (5) ensures that for each pair $(r;\xi)$ there is an $m < \infty$ such that $P\left(\prod_{i=1}^{m} Y_i > \xi/r\right) > 0$: with positive probability an increase in ξ will lead to a higher ratio R_n for n sufficiently large. Coupled with (8) and $\pi(r)$ strictly increasing in r, this fact enables us to conclude that $U(r;\xi)$ is strictly increasing in ξ. Q.E.D.

Proof of Theorem l : Because $R_n \leq \xi$ for all $n, U(r;\xi) \leq \pi(\xi)/(1-\beta)$. Therefore, $g(r) = U(r,r) < 1$. Next we seek a value R_0 of the initial ratio large enough to yield $g\left(R_0\right) > 1$. The idea of our proof is to select r_0 so large that the demand to capital ratio exceeds $\bar{r}$ with very high probability-and hence the one-period operating profit exceeds $1-\beta$-in each of the first N periods where N is large. With such a choice of R_0 the discounted expected operating profit associated with the first N periods would then be larger than 1 and, because N is large, the discounted expected operating profit associated with the remainder of the (infinite) planning period would be negligible. Let ξ be the smallest value of Y. If $\xi > 0$, then the choice $R_0 = \bar{r}/\xi^{N-1}$ ensures $R_{\|} \geq \bar{r}$ for all $n \leq N-1$. If $\xi = 0$, the more delicate argument presented below is required.

Define the nonincreasing function b by $h(t) = P(Y > t)$. It follows from (5) that $\lim_{t\to 0^+} b(t) = 1$. Consequently, there is a value $t > 0$ such that for all $r \geq T/t$

$$E\pi(Yr) > \pi(0)(1-b(t)) + \pi(\bar{r})b(t) \geq 1 - \beta + \delta,$$

where $2\delta = \pi(\bar{r}) - (1-\beta)$ and $\pi(0) \equiv \lim_{,\to 0}, \pi(r)$. Let $\varepsilon > 0$ be given and choose N such that $\beta^N \pi(0)/(1-\beta) > -\varepsilon/2$, and $1 - \beta^N > 1 - \varepsilon/2$.

Arguing as above and utilizing the independence of the $\{Y_i\}$, there is a value $\tau_N, 0 < \tau_N < 1$, such that if the initial ratio R_0 exceeds $\bar{r}/\tau_N$, then

$$E\pi\left[R_1 \prod_{t=1}^{''} Y_i\right] > \pi(0)\left(1 - h_n\left(\tau_N\right)\right) + \pi(\bar{r}) h_n\left(\tau_N\right) \geq 1 - \beta + \delta \quad n = 1, 2, \ldots, N-1$$

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Thus,

$$g\left(R_0\right)=U\left(R_0;R_0\right)>\sum_{n=0}^{N-1}(1-\beta+\delta)\beta^n-\varepsilon/2=1-\beta^N+\delta\left(1-\beta^N\right)/(1-\beta)-\varepsilon/2>1-\varepsilon.$$

Since $\varepsilon>0$ was arbitrary, this establishes the existence of a value R_0 with $g\left(R_0\right)\geq 1$. Coupling this fact and Lemma 1 proves that $g(\xi)>1$ for $\xi>R_0$.

When following a reservation ratio policy with parameter ξ, entry behaviour is memoryless in that it depends only upon the current ratio. Now suppose there is another memoryless competitive equilibrium, and let S be the associated set of ratios that induce entry. To say that entry occurs at the ratio s in S must signify a strictly positive amount of entry; this causes the ratio to fall, say to $s'<s$. In this case, the interval $(s',s]$ must be contained in S. In other words, S is the union of such intervals. If S is not an interval of the form (ξ,∞), then either $S=(a,b]$ for some $b<\infty$-which is clearly not an equilibrium-or there are disjoint intervals $(a,b]$ and $(c,d]$ in S with $a<b<c<d$ and a point α not in S with $b<\alpha<c$. Consider the discounted sum of the operating profits associated with entry from the initial ratio α and any given realization x $=\langle x_1,x_2,\ldots\rangle$ of the Markov process. It is strictly larger than the discounted sum of operating profits associated with the initial ratio b and the same realization x , for the demand to capital ratios are never smaller (and strictly larger in period 1) when the initial ratio is α. Taking the expected value over all realizations yields an expected discounted operating profit in excess of $U(b;S)=1$. Thus, entry is profitable from α so S is not an equilibrium. Q.E.D.

Proof of Corollary 1: If $\xi\leq\xi$, then

$$1=U(\xi\cdot;\xi\cdot)\geq U(\xi;\xi)>\hat{U}(\xi;\xi),$$

where the first inequality follows from Lemma 1 and the second inequality from $P(Y>1)>0,\pi(r)\leq\pi(r)$, and $\pi(r)<\pi(r)$ for some $r<\xi$. [If ξ were so small that $\hat{\pi}(r)=\pi(r)$ for all $r\leq\xi$, then $1=\hat{U}(\xi;\xi)=\hat{U}(\xi;\xi)<U(\xi\cdot;\xi\cdot)=1.]Thus, capitalenteringwhentheratiois\xi$ fails to earn its cost, a contradiction. Q.E.D.

Proof of Theorem 2: To begin, we shall show that if π is strictly concave, then $U(r;\xi,\sigma)$ is a strictly decreasing function of σ. In order to explicitly indicate the dependence of $U_n(r;\xi)$ and $U(r;\xi)$ on σ we write $U_n(\cdot;\xi,\sigma)$ and $U(\cdot;\xi,\sigma)$. We claim that if π is (strictly) convave, then $U_n(r;\xi,\sigma)$ and $U(r;\xi,\sigma)$ are (strictly) concave in r. The concavity of $U_n(\cdot;\xi,\sigma)$ follows from a trivial induction argument and the fact that the expected value of a concave function is itself concave; $U(r;\xi,\sigma)$ inherits this property as it is the limit of concave functions. The strict concavity of $U(r;\xi,\sigma)$ follows from that of π, (8), and the concavity of $U(r;\xi,\sigma)$. This verifies our claim.

We continue by recalling a well-known result. If Z_2 is riskier than Z_1 in the sense of second order stochastic dominance, written $Z_1>_2 Z_2$, and if f is nondecreasing and strictly concave, then

$$Ef\left(Z_1\right)>Ef\left(Z_2\right).$$

Fix ξ and r and notice that $\sigma_1<\sigma_2$ implies $Z_1>_2 Z_2$, where $Z_i=rY_{\sigma i}\wedge\xi$. By assumption π is strictly concave and nondecreasing so that

$$U_2\left(r;\xi,\sigma_1\right)=\pi(r)\beta E\pi\left(Z_1\right)>\pi(r)+\beta E\pi\left(Z_2\right)=U_2\left(r;\xi,\sigma_2\right).$$

Assume $U_n\left(\cdot;\xi,\sigma_1\right)>U_n\left(\cdot;\xi,\sigma_2\right)$. Then

$$U_{n+1}\left(r;\xi,\sigma_1\right)=\pi(r)\beta EU_n\left(Z_1;\xi;\sigma_1\right)>\pi(r)+\beta EU_n\left(Z_1;\xi,\sigma_2\right)\quad>\pi(r)+\beta EU_n\left(Z_2;\xi,\sigma_2\right)=U_{n+1}\left(r;\xi,\sigma_2\right.$$

where the first inequality follows from the induction hypothesis and the second inequality follows (A2) and $U_n\left(\cdot;\xi,\sigma_1\right)$ being nondecreasing and strictly concave (see the proof Lemma 1). This completes the induction argument. Passing to the limit on n, we can conclude that $U\left(r;\xi,\sigma_1\right)\geq U\left(r;\xi,\sigma_2\right)$. Applying (A2) to the strictly concave functions $U\left(\cdot;\xi,\sigma_i\right)$ yields

$$U\left(r;\xi,\sigma_1\right)=\pi(r)+\beta EU\left(Z_1;\xi;\sigma_1\right)\geq\pi(r)+\beta EU\left(Z_1;\xi,\sigma_2\right)\quad>\pi(r)+\beta EU_n\left(Z_2;\xi,\sigma_2\right)=U\left(r;\xi,\sigma_2\right)$$

so that $U(r;\xi,\sigma)$ is strictly decreasing in σ. Now fix $\sigma_1<\sigma_2$. From the definition of $\xi_{\sigma 1}$ and the fact that $U(r;\xi,\sigma)$ is strictly decreasing in σ, we have $1=U\left(\xi_{\sigma_1};\xi_{\sigma_1},\sigma_1\right)>U\left(\xi_{\sigma 1};\xi_{\sigma 1},\sigma_2\right)$. Moreover, Lemma 1 asserts that $U\left(\xi;\xi,\sigma_2\right)$ is a continuous strictly increasing function of ξ. Consequently, $U\left(\xi;\xi,\sigma_2\right)=1$ implies $\xi>\xi_{\sigma 1}$. Q.E.D.

Proof of Theorem 3: Fix $s<1$ and denote $\pi^{-1}(1-\beta)$ and $\pi^{-1}(s(1-\beta))$ by r_1 and r_s so $0<r_s<r_1$. (Reference to Figure 2 facilitates the proof.) To begin, consider the function Λ implicitly defined by

$$V(\Lambda(L);L,\Lambda(L))=1\quad\text{for}\quad 0\leq L\leq r_1.$$

The content of (A3) is that with each exit level L the function Λ associates an entry level $\Lambda(L)$ such that the two levels policy $(L,\Lambda(L))$ satisfies (13): the net return to entry is non-negative if and only if the demand to capital ratio (equals or) exceeds $\Lambda(L)$. There is no reason to expect that exit is induced if and only if the ratio is less than $L:(L,\Lambda(L))$ need not be a competitive equilibrium.

Similarly, consider the function λ implicitly defined by

$$V(L;L,\lambda(L))=s\quad\text{for}\quad\alpha<L\leq r_s$$

Given the exit level L, (A4) asserts that λ selects an entry level $\lambda(L)$ satisfying (14). Capital prefers to exit if and only if the ratio is less than L. Because the expected discounted value $V(L;L,\infty)$ of the operating profits can be less than s for sufficiently small positive values of L, there may be no solution $\lambda(L)$ of (A4) when L is close to zero; consequently, $\lambda(L)$ is defined on the interval $(\alpha,r_s]$, where $\alpha\geq 0$ is unspecified.

It is important to note that $\Lambda\left(r_1\right)=r_1,\Lambda$ is strictly decreasing, and Λ is continuous. (Of related interest is $\Lambda(0)=\xi$.) Furthermore, $\lambda\left(r_s\right)=r_s,\lambda$ is strictly decreasing, and λ is continuous.

Figure 2 illustrates the definitions of Λ and λ and reveals an obvious but important step in the proof: the existence of a two levels policy inducing a competitive equilibrium is equivalent to the existence of a value $L\geq 0$ such that

$$\Lambda(L)=\lambda(L).$$

The major step in demonstrating the existence of a solution to (A5) is to show there is an L such that $\lambda(L) > \Lambda(L)$. Because λ and Λ are continuous and Λ is strictly decreasing, it suffices to show $\lambda(L) = \max\{\Lambda(x) : 0 \leq x \leq r_1\} = \Lambda(0)$. Toward this goal, define u by

$$v(L) = V(L; L, \Lambda(0)) \quad \text{for} \quad 0 \leq L \leq r_{\mathrm{s}}$$

so that $v(L) = s$ if and only if $\lambda(L) = \Lambda(0)$. The feasibility condition (15) yields

$$\boldsymbol{v}(0) = V(0; 0, \Lambda(0)) = \pi(0)/(1-\beta) < s$$

while $V(r; L, H)$ strictly increasing in H yields

$$v\left(r_s\right) = V\left(r_s; r_s, \Lambda(0)\right) > V\left(r_s; r_s, r_s\right) = s$$

Linking (A7), (A8), and the continuity of v produces a value of L, label it l, such that $\boldsymbol{v}(l) = s$ and $0 < l < r_s$:

$$\lambda(l) = \Lambda(0) > \Lambda(l).$$

The desired result, the existence of a solution to (A5), label it L_s, follows immediately from (A9) and the continuity of λ.

Demonstrating $\left(L_s, \Lambda\left(L_s\right)\right) \equiv \left(L_s, H_s\right)$ is the unique two levels policy which is a competitive equilibrium follows immediately from verifying

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$$\Lambda'(L) > \lambda'(L), \quad \text{for all } \alpha < L < r_s.$$

Abbreviating $V_i(L; L, \lambda(L))$ and $V_i(\Lambda(L); L, \Lambda(L))$ by $V_i(L)$ and $V_i(\Lambda(L))$, respectively, where $V_i(r) = \partial V(r; L, H)/\partial i, i = r, L, H$, implicit differentiation of (A3) and (A4) yields

$$-\Lambda'(L) = V_1(\Lambda(L))/\left[V_H(\Lambda(L)) + V_H(\Lambda(L))\right]$$

and

$$-\lambda'(L) = \left[V_r(L) + V_L(L)\right]/V_H(L).$$

Rearranging (A11) reveals the equivalence between (A10) and (A12):

$$\frac{V_r(L) + V_L(L)}{V_L(\Lambda(L))} \bullet \frac{V_r(\Lambda(L)) + V_H(\Lambda(L))}{V_H(L)} > 1.$$

It is enough to show that each ratio in (A12) exceeds 1 . This is effected by verifying

$$[V(L+\varepsilon; L+\varepsilon, H) - V(L; L, H)] - [V(H; L+\varepsilon, H) - V(H; L, H)] > 0$$

and

$$[V(H+\varepsilon; L, H+\varepsilon) - V(H; L, H)] - [V(L; L, H+\varepsilon) - V(L; L, H)] > 0,$$

where $\varepsilon > 0, \alpha < L < r_s$ and $L + \varepsilon < H$. To begin fix L, ε, and H and take any sample point ω in the probability space Ω. Let $b_n(\omega)$ denote the demand to capital ratio at time

n when starting from the ratio H at time zero, demand levels are determined by ω, and entry and exit is determined by the two levels policy (L, H). Similarly, let $b_n^+(\omega)$ be this ratio when ($L+\varepsilon, H$) is the two levels policy. Also, let $l_n(\omega)\,[l_n^+(\omega)]$ denote the ratio at time n when starting from the ratio $L[L+\varepsilon]$ at time zero, demand levels are determined by ω, and entry and exit is determined by the two levels policy $(L, H)[(L+\varepsilon, H)]$. Of course, $l_n^+(\omega) \geq l_n(\omega), b_n^+(\omega) \geq h_n(\omega)$, and $h_n(\omega) \geq l_n(\omega)$.

Let τ be the first time the 'upper pair' $b_n^+$and b_n separates:

$$\tau(\omega) = \min\left\{n : {h_n}^+(\omega) - h_n(\omega) > 0\right\}.$$

Observe not only that $b_r^+(\omega) = l_r^+(\omega)$ but also

$$h_n^+(\omega) = l_n^+(\omega) \quad \text{for all} \quad n \geq \tau(\omega).$$

Using (A14) and (A15), verifying (A13a) is straightforward:

$$[V(L+\varepsilon; L+\varepsilon, H) - V(L; L, H)] - [V(H; L+\varepsilon, H) - V(H; L, H)]$$

$$= \int_\Omega \sum_{n=0}^{\infty} \beta^n \left\{\left[\pi\left(l_n^+(\omega)\right) - \pi\left(l_n(\omega)\right)\right] - \left\{\pi\left(b_n^+(\omega)\right) - \pi\left(h_n(\omega)\right)\right\}\right\} P(d\omega) \quad = \int_\Omega \left\{\sum_{n=0}^{\tau-1} \beta^n \left[\pi\left(l_n^+(\omega)\right) - \pi\left(l_n\right.\right.\right.$$

Verification of (A13b) proceeds in the same manner. Let $b_n^+(\omega)$ and $l_n^+(\omega)\,[b_n(\omega)$ and $l_n(\omega)]$ denote the ratios at time n when starting from the initial ratio $H+\varepsilon$ and $L[H$ and $L]$, respectively, demand levels are determined by ω, and capital movements are determined by the two levels policy ($L, H+\varepsilon$) [(L, H)]. Let τ be the first time the 'lower pair' $l_n^+$and l_n separates:

$$\tau(\omega) = \min\left\{n : l_n^+(\omega) - l_n(\omega) > 0\right\}.$$

As before $l_n^+(\omega) \geq l_n(\omega)$ and $h_n^+(\omega) \geq h_n(\omega)$, but (A15) is replaced by

$$b_n(\omega) = l_n(\omega) \text{ and } {b_n}^+(\omega) \geq {l_n}^+(\omega) \text{ for all } n \geq \tau(\omega).$$

The same argument reveals that (A136) easily follows from (A16) and (A17). We prove by contradiction that this is the unique memoryless equilibrium. Suppose there is another memoryless competitive equilibrium; let E^+ and E^-be the set of ratios at which entry and exit occur, respectively.

Given $e^+ \in E^+$and $e^- \in E^-, e^- < e^+$necessarily obtains. Otherwise, $e^- = e^+$or $e^- > e^+$. If $e^- = e^+$, capital stands ready to enter at a cost of 1 and immediately exit in return for the salvage value $s < 1$, an impossibility. If $e^- > e^+$, let $R_n^+$and R_n^-be the ratios at time n associated with the alleged competitive equilibrium when starting from $e^+$and e^-, respectively. Because $E^+ \cap E^- = \emptyset$ as shown above, $R_n^- \geq R_n^+$for all $n \geq 0$ in which case

$$1 >_s = E\sum_{n=0}^{\infty} \beta^n \pi\left(R_n^-\right) \geq E\sum_{n=0}^{\infty} \beta^n \pi\left(R_n^+\right) = 1.$$

The argument presented in the proof of Theorem 1 can be used to show E^+ and $E^-$are each connecred sers, say $E^+ = (e^+, b)$ and $E^- = [a, e^-)$with $e^- < e^+$. Proving $a = 0$ and $b = \infty$

is straightforward. Consequently, this other memoryless competitive equilibrium is a two levels policy.

Proof of Theorem 4: We begin by explicitly noting the dependence on s of the function λ, defined in (A4), by writing $\lambda(L;5)$ so (A4) becomes

$$V(L;L,\lambda(L;s)) = s.$$

Given the exit level L, (A18) asserts that $\lambda(\cdot;s)$ selects an entry level $\lambda(L;s)$ such that the two levels policy ($L,\lambda(L;s)$) satisfies (14). Recall that $V(r;L,H)$

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is the expected discounted operating profit associated with a unit of capital which never exits. We claim $\lambda\left(L;s'\right) > \lambda(L;s)$ for $s' > s$. If not, then an increase in s to s' will induce capital to exit at the ratio L and earn s' rather than remain in the industry and only earn $V\left(L;L,\lambda\left(L;s'\right)\right) \leq V(L;L,\lambda(L;s)) = s$.

Similarly, replace $\Lambda(L)$ by $\Lambda(L;s)$ so that, as per (A3),

$$V(\Lambda(L;s);L,\Lambda(L;s)) = 1$$

Given L, an increase in s has no bearing on the value of $V(r;L,H)$; accordingly, an increase in s leaves $\Lambda(L;s)$ unchanged.

As $\lambda(L;s) > \Lambda(L;s)$ and $\lambda(\cdot;s)$ and $\Lambda(\cdot;s)$ are both decreasing, L_s, the value of L such that $\Lambda(L;s) = \lambda(L;s)$, must be increasing in s. As per (13), an increase in L_s necessitates a decrease in H_s. Q.E.D.

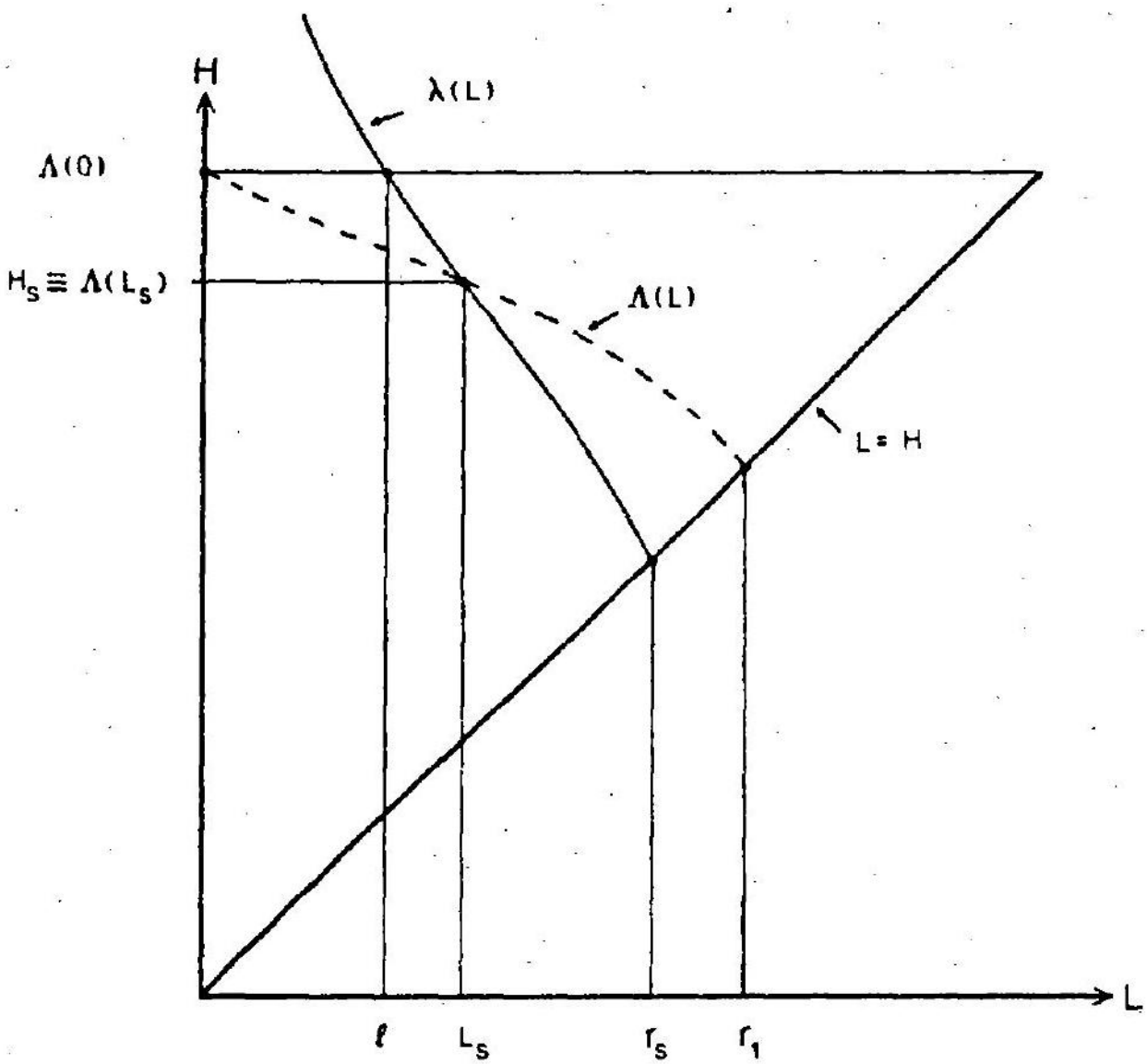

Figure 2 Functions Used in Proof of Theorem 3